Effects of peppermint teas on plasma testosterone, follicle-stimulating hormone, and luteinizing hormone levels and testicular tissue in rats.

❖ ❖ ❖

OBJECTIVES: To justify the effects of Mentha piperita labiatae and Mentha spicata labiatae herbal teas on plasma total testosterone, luteinizing hormone, and follicle-stimulating hormone levels and testicular histologic features. We performed this study because of major complaints in our area from men about the adverse effects of these herbs on male reproductive function. METHODS: The experimental study included 48 male Wistar albino rats (body weight 200 to 250 g). The rats were randomized into four groups of 12 rats each. The control group was given commercial drinking water, and the experimental groups were given 20 g/L M. piperita tea, 20 g/L M. spicata tea, or 40 g/L M. spicata tea. RESULTS: The follicle-stimulating hormone and luteinizing hormone levels had increased and total testosterone levels had decreased in the experimental groups compared with the control group; the differences were statistically significant. Also, the Johnsen testicular biopsy scores were significantly different statistically between the experimental groups and the control group. Although the mean seminiferous tubular diameter of the experimental groups was relatively greater than in the control group, the difference was not statistically significant. The only effects of M. piperita on testicular tissue was segmental maturation arrest in the seminiferous tubules; however, the effects of M. spicata extended from maturation arrest to diffuse germ cell aplasia in relation to the dose. CONCLUSIONS: Despite the beneficial effects of M. piperita and M. spicata in digestion, we should also be aware of the toxic effects when the herbs are not used in the recommended fashion or at the recommended dose.